NBA 2K20 Player Rating Predictions Using Machine Learning and Ensemble Learning Approaches

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Abstract. The sports industry has witnessed significant growth, capturing the attention of a diverse audience. Within this burgeoning industry, athletes hold a substantial share, and their performances and ratings assume paramount significance, particularly in the determination of their remuneration. This study delves into the realm of sports video games, focusing on the NBA 2K20 official video game, employing regression analysis and various Machine Learning (ML) algorithms, including k-Nearest Neighbors (Knn), Lasso, and Extreme Gradient Boosting (XGBoost), to evaluate player ratings. Moreover, the study investigates the impact of ensemble methods on model performance. In this study, we applied ML algorithms within the domain of sports video games. This endeavour not only yielded valuable insights into the gaming experience but also highlighted the potential for data-driven decision-making within the sports industry. This study shows that Knn consistently exhibits superior predictive accuracy in both paradigms. While Boosting showed improvement, Bagging emerged as particularly prominent, engendering incremental enhancements in MAE and R2 values across all models, highlighting its effectiveness in ameliorating predictive precision.

Keywords: Machine Learning · Bagging · Boosting · NBA 2K20 · Video Games

1 Introduction

In sports science, analysis of players' data is important for managers in order to correctly manage players [1]. Managers observed players in order to obtain detailed information about them during the training in the past. Then, they made a plan for each player in terms of correctly managing his/her improvement for the next/following training. However, people making observations during the data collection process may make mistakes [2]. In other words, the data obtained by the managers as a result of observation and the

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analysis of these data may not be at the desired level. In this sense, players may not manage correctly in terms of their improvement, and so it can be considered an important problem in sports science. On the other hand, the use of Artificial Intelligence (AI), Machine Learning (ML), and Deep Learning (DL) algorithms in terms of analysing data about players' performance has significantly increased with the development of technology especially in the last decade [3].

Data analysis have been carried out employing ML algorithms in nearly all sport science including basketball, football, volleyball, handball, etc. It should be also highlighted that ML learning algorithms mostly preferred to analyse online game data besides a variety of sports sciences.

Qader et al. [4], conducted a study wherein they utilized Artificial Neural Networks (ANN) to train the background of the DAMA game. They implemented the NEAT algorithm, coupled with extensive training time involving ANN, which yielded promising results. This utilization of ML algorithms empowers managers by allowing them to focus on detailed player data and a plethora of features.

The interdisciplinary nature of sports science, which includes diverse areas like basketball and football can be examined as a research topic. For instance, one study applied machine learning methods to FIFA19 soccer video game for forecasting player ratings, using dataset from Kaggle. The researchers tested different feature selection techniques like ReliefF, Gain Ratio, Chi-Square, and Information Gain for better performance. Buyrukoğlu and Savaş conducted a machine learning approach to predict FIFA19 player ratings. Their work focuses on how machine learning can be applied to predict performance metrics for athletes and sports video game characters [5]. Notably, the researchers employed a sophisticated stacked ensemble model, incorporating Level 0 models, including Deep Neural Network (DNN), Random Forest (RF), and Gradient Boosting (GB), and a Level 1 model, Logistic Regression (LR). Impressively, by employing the Chi-Square feature selection method in conjunction with the proposed stacked model, a commendable accuracy rate of 83.9% was achieved. While football undoubtedly enjoys widespread recognition as one of the most prominent subjects within the realm of sports sciences, it is essential to acknowledge that basketball holds a similar stature, as recognized by sports scientists [6, 7]. Consequently, the subsequent paragraph will expound upon existing scholarly contributions that delve into the examination of basketball-related data through the prism of ML algorithms.

In a study by Zaid and Rajendran [8], the accuracy of ML algorithms in predicting adult income was explored. The study employed the "Adult income dataset," a publicly available resource on Kaggle. The Decision Tree (DT) and Gaussian Naïve Bayes (GNB) algorithms were the principal ML approaches examined. Notably, the DT model exhibited a higher accuracy rate at approximately 84%, outperforming the GNB model, which achieved a 79% accuracy rate. However, a noteworthy limitation of this study is that it predicted income class rather than actual salary.

Asadi and Tasdemir [9], aimed to predict the market value of football players using ML techniques. The dataset utilized in this investigation was sourced from the publicly available FIFA20 video game dataset on Kaggle. The RF regression model yielded the highest R2 score of 95%, while other regression methods, including LR, Multiple Linear Regression (MLR), and RT, were also employed.

Jain et al. [10], conducted a study focused on predicting the salaries of NBA players using ML methodologies. They compiled a private dataset by collecting data from the 2020-2021 season from ESPN and NBA official websites. Among various regression models, the RF regressor achieved the highest R2 result of 88.18%, surpassing Multiple Linear Regression (MLR), DT, and Extreme Gradient Boosting (XGB).

Li [11], proposed a study to create a salary regression model for NBA players. This study primarily relied on empirical economic models to analyze detailed parameters related to NBA player salaries. However, a notable limitation of this study is the absence of advanced systems such as ML or DL.

Horvat et al. [12], applied ML algorithms to predict the outcomes of NBA games. This extensive study analyzed data from 6567 NBA basketball games spanning five seasons, from 2013/2014 through 2017/2018. Data collection was facilitated by a web scraping program to obtain statistics from the official NBA website, and analysis was conducted using the Basketball Coach Assistant (BCA) information system. Feature selection was manually performed to optimize results, with K-Nearest Neighbors (KNN) achieving the highest accuracy of 57.9% compared to LR, Naïve Bayes (NB), DT, Multilayer Perceptron Layer (MLP), RF, and LogitBoost.

Schanilec et al. [13], conducted a study to predict MVP candidates for an NBA season using public NBA data and four distinct ML algorithms: Support Vector Machine (SVM), Naïve Bayes (NB), Random Forest (RF), and Gradient Tree Boosting (GTB). All of the ML models attained accuracy scores exceeding 99%.

Thabtah et al. [14], aimed to predict NBA game results through ML methods using a public dataset available on Kaggle, which contained player statistics spanning 30 years. Feature selection techniques, including Multiple Regression, Correlation Feature Set (CFS), and RIPPER algorithm, were applied. Three different models, NB, Artificial Neural Network (ANN), and Logistical Model Tree (LMT), achieved accuracy rates of 80%, 80%, and 83%, respectively, compared to other feature sets.

Zuo et al. [15], proposed a model for predicting player ratings based on a study that utilized publicly available datasets from Amazon reviews. This model employed a correlation-based regression approach to evaluate prices in categories such as pet supplies, compact discs, office products, and video games.

Nor et al. [16], conducted a study to predict user ratings in video games, focusing on visualization and prediction processes using web analytics. Notably, this study exclusively utilized the RF classification method, achieving an accuracy of 96.12%.

Karaçam et al. [17], explored the mental state and emotion recognition of referees in basketball games. Data collection was conducted using the Warwick-Edinburgh Mental Well-Being Scale. However, a significant limitation of this study is its reliance on the authors' subjective opinions, with no application of data processing methods such as ML.

The aforementioned studies provided convincing scores based on their purposes. However, to the best of the authors' knowledge, researchers did not focus on comparing the performance of the bagging and boosting ensemble approach in the prediction of basketball players' overall ranking grades. Thus, the objectives of this study are listed below with bullet points according to the limitations of the studies.

- To determine the optimal dataset by applying data preprocessing techniques.

- To make a prediction of basketball players' overall ranking score by employing Knn, Lasso, XGBoost, Bagging, and Boosting Ensemble algorithms.
- To compare the performance of the employed Bagging and Boosting Ensemble approaches.

This study contributes the determination of the player's marketing value correctly based on these objectives.

The structure of this manuscript is organized as follows: The proposed basketball players' overall ranking prediction approach is presented in Sect. 2. Detailed information about the results obtained from employed ML algorithms and a discussion of the results are available in Sect. 3. Conclusion and further studies are shown in the final section.

2 Materials and Method

In this section we provide a concise overview of our research framework, as illustrated in Fig. 1. We delve into the components, techniques, and processes involved in developing and assessing the transparency of our ML model. Additionally, we emphasize the critical role of tools, data sources, and analytical methods in laying the groundwork for presenting our findings and results in subsequent sections.

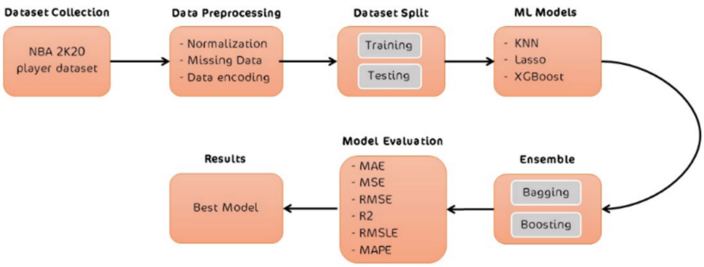

Fig. 1. The proposed ML structure

2.1 Data Collection

The dataset was used from Kaggle, a highly trusted data repository known for its reliability. Specifically, the dataset used is the "NBA 2k20 player dataset", which includes comprehensive information on 429 current NBA players, including their full names, countries of origin, and nationalities [18]. Kaggle is a trusted database provider that strict data verification procedures to ensure the reliability of the dataset for data mining applications. Some of the features like player names, nationality, and game statistics in NBA 2K20 dataset provide a solid foundation for investigating player attributes and performance metrics in a predictive model. Selection of the data is an import metric for research. Results will be better with a well-prepared dataset [19].

2.2 Data Preprocessing

The data preprocessing has some important data preparation procedures. Firstly, normalizing numerical variables to standardize scales is the most important part of the preprocessing. Imputing missing values by taking means, and encoding categorical features numerically are the other important specifications of a dataset. These steps improve data quality. Preprocessing procedures make the dataset suitable for later analysis and modeling. This process is especially important for features with different units or scales and for dealing with missing real-world data [20].

2.3 ML Models

Machine learning algorithms specialize in modeling relationships between variables to predict continuous outcomes. However, each algorithm has unique strengths and limitations. Researchers must carefully consider the right model for a given problem and dataset. As a result, advancing and extending ML to make it applicable across diverse industries and domains is crucial [21].

Classical statistical methods rely on assumptions like normality and equal variance that are often violated in practice. Machine learning algorithms present an alternative approach in these situations. Machine learning methods focus on making optimal predictions from the available data without losing information or imposing restrictions.

K-nearest neighbors (KNN) is a versatile algorithm for classification and regression. It predicts based on proximity to other data points but requires careful selection of K and distance metric. While computationally demanding for large datasets, KNN remains valuable for its simplicity and interpretability [22].

Lasso regression is known for its feature selection and regularization capabilities. By encouraging sparse coefficient estimates, Lasso handles datasets with many features, preventing overfitting and improving model interpretability across domains [23].

XGBoost uses an ensemble of decision trees to enhance accuracy. It leverages gradient descent optimization for speed, regularization to control overfitting, and provides feature importance scores for interpretability. Highly effective at capturing complex data patterns, handling missing values, and diverse data types [24].

2.4 Ensemble Methods

Ensemble methods like Boosting and Bagging are crucial machine learning techniques, improving predictive performance and model robustness [25]. Bagging trains models on random data subsets then combines predictions, reducing variance and overfitting [26]. Boosting iteratively trains base learners to enhance better models for improving accuracy of the system [27]. These methods leads to highly accurate ensemble classifiers with reduced bias. Ensemble methods exemplifies the "wisdom of crowds" concept, with combined models often outpacing individual ones, making them vital in machine learning.

2.5 Evaluation Metrics

Assessing performance of a regression model is essential part of the predictive applications., Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), R-squared (R^2), Root Mean Squared Logarithmic Error (RMSLE), and Mean Absolute Percentage Error (MAPE) are some of the crucial metrics of a regression model [28]. MAE measures average absolute differences between predictions and actuals, indicating accuracy. MSE and RMSE account for squared and squared-rooted differences, emphasizing outlier impact. R^2 quantifies the explained variance in the target variable, while RMSLE is useful for logarithmic transformations and underestimation sensitivity. MAPE calculates prediction errors as a percentage of actual values, offering accuracy insights. These metrics collectively guide model selection for specific tasks and datasets, providing a comprehensive performance assessment.

3 Results and Discussion

In this section, we present the results of suggested ML models, both individually and in ensemble configurations. The performance metrics, including MAE, MSE, RMSE, R^2, RMSLE, and MAPE, are tabulated for each model and ensemble method. Out dataset was used 70:30 for training and testing consequently. The following subsections provide detailed performance results and are referred to in the subsequent discussion.

3.1 Performance of Individual ML Models

In Table 1, an extensive evaluation of the performance of the suggested three distinct ML models is presented. The discernible trend within the results is the superior performance of the Knn model across multiple performance metrics. Specifically, Knn attains the lowest MAE at 3.3017, signifying its proficiency in generating predictions characterized by minimal average absolute errors. Furthermore, Knn prominently exhibits the highest R^2 value, registering at 0.5378. This R^2 value signifies an admirable model fit to the data, denoting a considerable extent of variance explained by the model. Nevertheless, it is noteworthy that Knn does display a relatively elevated RMSLE of 0.0544, indicating a potential variability in its performance when applied to different data instances.

Lasso and XGBoost, while providing competitive results, show slightly higher errors and lower R^2 values compared to Knn. This implies that Knn may be better suited for this specific prediction task based on individual model performance.

Table 1. Performance of Individual ML Models

ML Model	MAE	MSE	RMSE	R^2	RMSLE	MAPE
Knn	3.3017	20.9553	4.4475	0.5378	0.0544	0.0412
Lasso	4.1044	25.9807	5.0099	0.4166	0.0628	0.0528
XGBoost	4.3781	30.4268	5.4491	0.3041	0.0688	0.0570

3.2 ML Model Performance Using Bagging Ensemble

Table 2 displays the outcomes when the identical ML models are employed in conjunction with the Bagging ensemble technique. Bagging aims to reduce variance by aggregating multiple instances of a model trained on different subsets of the data. Knn continues to perform better in this ensemble model, with a minimal MAE of 3.2760 and a maximum R^2 score of 0.5398 between the models. It's interesting to see that using bagging has slightly improved each model's performance, especially when it comes to raising R^2 values and decreasing MAE.

These findings imply that Bagging is useful in raising the models' prediction accuracy, with Knn continuing to be the best option.

Table 2. ML Model Performance Using Bagging Ensemble

ML Model	MAE	MSE	RMSE	R^2	RMSLE	MAPE
Knn	3.2760	20.9054	4.4344	0.5398	0.0542	0.0408
Lasso	4.1031	26.0101	5.0127	0.4160	0.0628	0.0528
XGBoost	4.3579	29.9281	5.3983	0.3181	0.0682	0.0567

3.3 ML Model Performance Using Boosting Ensemble

Table 3 showcases the results obtained when employing the identical ML models in conjunction with the Boosting ensemble methodology.

Unlike Bagging, Boosting focuses on reducing bias and improving model accuracy by iteratively giving more weight to misclassified instances. In this scenario, Knn maintains its lead in terms of MAE of 3.2751 and R^2 of 0.5374 compared to Lasso and XGBoost. However, it is noteworthy that the Boosting ensemble meth-od does not exhibit the same level of improvement observed with Bagging.

These results indicate that while Boosting can enhance the performance of the models, its impact is less pronounced than that of Bagging in the context of our prediction task. Knn still stands out as the most favorable model in terms of pre-dictive accuracy.

Table 3. ML Model Performance Using Boosting Ensemble

ML Model	MAE	MSE	RMSE	R2	RMSLE	MAPE
Knn	3.2751	21.0004	4.4404	0.5374	0.0548	0.0413
Lasso	4.1221	26.6630	5.0836	0.3964	0.0641	0.0535
XGBoost	4.3039	29.2126	5.3317	0.3350	0.0673	0.0560

The results presented in this section illustrate the performance of individual machine learning models as well as ensemble methods that combine multiple models. The k-nearest neighbors algorithm consistently exhibits superior predictive power, whether used by itself or within an ensemble method. This indicates that KNN is a viable and dependable choice for the particular prediction problem under investigation. When selecting the final model to deploy, computational efficiency and interpretability are critical additional factors to consider alongside accuracy. To further boost the predictive capabilities of the models, future research could explore tuning the models' hyperparameters and testing other ensemble approaches to amalgamate the strengths of multiple models. Moreover, domain-specific insights and feature engineering may provide opportunities to improve model performance and gain a deeper understanding of the underlying data patterns.

4 Conclusion and Future Works

In this manuscript, insights into the gaming experience and player performance in NBA 2K20 were sought by predicting basketball players' overall ranking scores in the study. Utilizing Knn, Lasso, XGBoost, Bagging, and Boosting Ensemble algorithms, patterns and relationships within the dataset were aimed to be uncovered, thereby offering specific insights into players' in-game performance.

Moreover, this study delved into the impact of employing ensemble methodologies on model performance. In both ensemble paradigms, Knn consistently exhibited superior predictive accuracy. Notably, Bagging engendered incremental enhancements in MAE and R2 values across all models.

While Boosting did elicit some degree of improvement in model performance, it fell short of achieving the commensurate level of enhancement observed with Bagging. These findings underscore the efficacy of ensemble techniques, with Bagging particularly prominent, in ameliorating the predictive precision of the models for the specified prediction task. Nonetheless, Knn consistently emerged as the most judicious choice, whether as an individual model or as an integral component within ensemble frameworks.

The research advocates for additional exploration of alternative ensemble methods, refinement of hyperparameters, and the application of domain specific expertise to enhance model performance and gain deeper insights. The conclusion emphasizes the importance for practitioners to weigh interpretability and computational efficiency alongside predictive accuracy when selecting models for real-world applications.

The study recommends applying ensemble techniques, particularly Bagging, in real-world sports for improved player rating predictions. This enhances accuracy in strategic decisions about team composition, player recruitment, and performance evaluation. Prioritizing interpretability and computational efficiency ensures practicality in dynamic sports environments, enhancing decision-making.

Future research should explore alternative ensemble techniques and fine-tune hyperparameters for better model performance. Domain-specific expertise and advanced feature engineering can provide deeper insights into data patterns and improve predictions. When selecting a model for real-world use, consider not only predictive accuracy but also interpretability and computational efficiency.

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